

AWS Task-3

Launch 2 EC2 instances, install a web server on both, add both instances to an Application Load Balancer, and when the Load Balancer DNS name is accessed from the browser, traffic should go to either EC2 server.

Step-1: Launch Two EC2 Instances

- Log in to the AWS Management Console, go to EC2, click Launch Instance, name it like Web-Server-1, and set Number of instances = 2.
- Choose Amazon Linux, select t2.micro, and avoid a key pair for this task.
- In Network Settings, enable Auto-assign public IP and create a security group allowing HTTP port 80 from 0.0.0.0/0 and SSH port 22 from My IP.
- In Advanced Details, under User data, paste a startup script to install and start the web server so the instances can be tested through the load balancer later.

```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo "<h1>Hello from $(hostname -f)</h1>" >
/var/www/html/index.html
```

Instance summary for i-049d361536df152b6 (webserver1) info

Updated less than a minute ago

Instance ID
i-049d361536df152b6

IPv6 address
-

Hostname type
IP name: ip-172-31-32-119.ap-south-1.compute.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
13.206.185.194 [Public IP]

IAM role
-

IMDSv2
Required

Operator
-

Public IPv4 address
13.206.185.194 [open address ↗](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-32-119.ap-south-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-067c73cc11a5305e9 [↗](#)

Subnet ID
subnet-0b42562fcb408bac [↗](#)

Instance ARN
arn:aws:ec2:ap-south-1:314727362308:instance/i-049d361536df152b6

Private IPv4 addresses
172.31.32.119

Public DNS
ec2-13-206-185-194.ap-south-1.compute.amazonaws.com [open address ↗](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
[Opt-in to AWS Compute Optimizer for recommendations.](#) | [Learn more ↗](#)

Auto Scaling Group name
-

Managed
false

Instance summary for i-0503d3360a8c6d69d (webserver2) [Info](#)

Updated less than a minute ago

Instance ID

[i-0503d3360a8c6d69d](#)

IPv6 address

–

Hostname type

IP name: ip-172-31-32-74.ap-south-1.compute.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

[43.205.213.154](#) [Public IP]

IAM role

–

IMDSv2

Required

Operator

–

Public IPv4 address

[43.205.213.154](#) | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

[ip-172-31-32-74.ap-south-1.compute.internal](#)

Instance type

t3.micro

VPC ID

[vpc-067c73cc11a5305e9](#)

Subnet ID

[subnet-0b42562fcb408bac](#)

Instance ARN

[arn:aws:ec2:ap-south-1:314727362308:instance/i-0503d3360a8c6d69d](#)

Private IPv4 addresses

[172.31.32.74](#)

Public DNS

[ec2-43-205-213-154.ap-south-1.compute.amazonaws.com](#) | [open address](#)

Elastic IP addresses

–

AWS Compute Optimizer finding

[Opt-in to AWS Compute Optimizer for recommendations.](#) | [Learn more](#)

Auto Scaling Group name

–

Managed

false

[EC2](#) > [Security Groups](#) > [sg-01cf7047d6368c78 - launch-wizard-2](#) > Edit inbound rules

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID

sg-0cd5a5c067da986a4

Type

[Info](#)

HTTP

Protocol

[Info](#)

TCP

Port range

[Info](#)

80

Source

[Info](#)

Custom

Description - optional

[Info](#)

[Delete](#)

sg-00e90c1a9fb2574de

SSH

TCP

22

Custom

[0.0.0.0/0](#) [X](#)

[0.0.0.0/0](#) [X](#)

[Delete](#)

[Add rule](#)

Step-2: Create a Target Group

- Before creating the load balancer, create a Target Group by navigating to Target Groups under EC2 dashboard, choose Instances, name it web-server-target-group, and set HTTP, Port 80.
- Then select the two EC2 instances, click Include as pending below, and click Create target group.

Review and create

Review your target group configuration before creating

Step 1: Target group details

[Edit](#)

Target group details

Name web-server-target-group	Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1
VPC vpc-067c73cc11a5305e9 i	IP address type IPv4		

Health check details

Health check protocol HTTP	Health check path /	Health check port traffic-port	Interval 30 seconds
Timeout 5 seconds	Healthy threshold 5	Unhealthy threshold 2	Success codes 200

Step 2: Register targets

[Edit](#)

Targets (2)

Instance ID	Name	Port	Zone
i-049d361536df152b6 i	webserver	80	ap-south-1a
i-0503d3360a8c6d69d i	webserver	80	ap-south-1a

Step 3: Create the Application Load Balancer (ALB)

- In EC2 dashboard, click Load Balancers, click Create load balancer, choose Application Load Balancer, name it like my-web-server-lb, set it as Internet-facing, and select the VPC with at least two Availability Zones.
- Choose a security group allowing HTTP port 80 from the internet, route the HTTP:80 listener to the target group, and click Create load balancer.

my-web-server-lb [🔄](#) [Actions](#)

▼ Details

Load balancer type
Application

Scheme
Internet-facing

Status
🟢 Active

Hosted zone
ZP97RAFLXTNZK

VPC
[vpc-067c73cc11a5305e9](#) [i](#)

Load balancer IP address type
IPv4

Availability Zones
[subnet-0b2382857ccd0774b](#) [i](#) ap-south-1b (aps1-az3)
[subnet-0b42562fcb408bac](#) [i](#) ap-south-1a (aps1-az1)

Date created
April 25, 2026, 23:57 (UTC+05:30)

Load balancer ARN
[arn:aws:elasticloadbalancing:ap-south-1:314727362308:loadbalancer/app/my-web-server-lb/3bf8185bc78912b0](#)

DNS name [Info](#)
[my-web-server-lb-1352965561.ap-south-1.elb.amazonaws.com](#) (A Record)

[Listeners and rules](#)

[Network mapping](#)

[Resource map](#)

[Security](#)

[Monitoring](#)

[Integrations](#)

[Attributes](#)

[Capacity](#)

[Tags](#)

Listeners and rules (1) [Info](#)

[🔄](#) [Manage rules](#) [Manage listener](#) [Add listener](#)

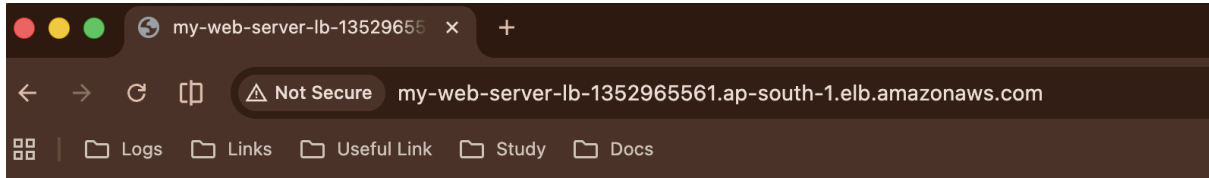
A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

[Filter listeners](#)

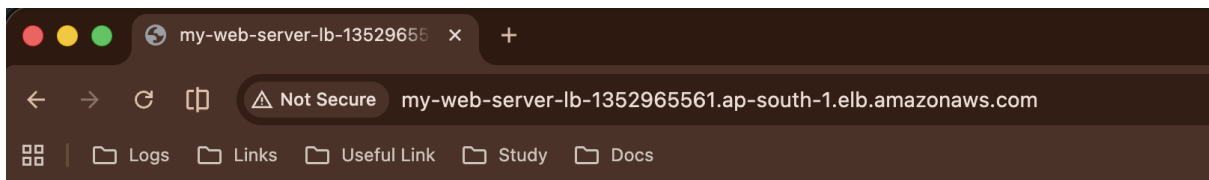
☐	Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS
☐	HTTP:80	• Forward to target group web-server-target-group i: 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable

Step 4: Verification

- Once the Load Balancer status becomes Active and both instances in the Target Group show Healthy, copy the Load Balancer DNS name, open it in the browser, and refresh the page to see the “Hello from...” message switch between the two EC2 instances.



Hello from ip-172-31-32-119.ap-south-1.compute.internal



Hello from ip-172-31-32-74.ap-south-1.compute.internal